MECHANISM & NEURAL CONTROL OF DEGLUTITION

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Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

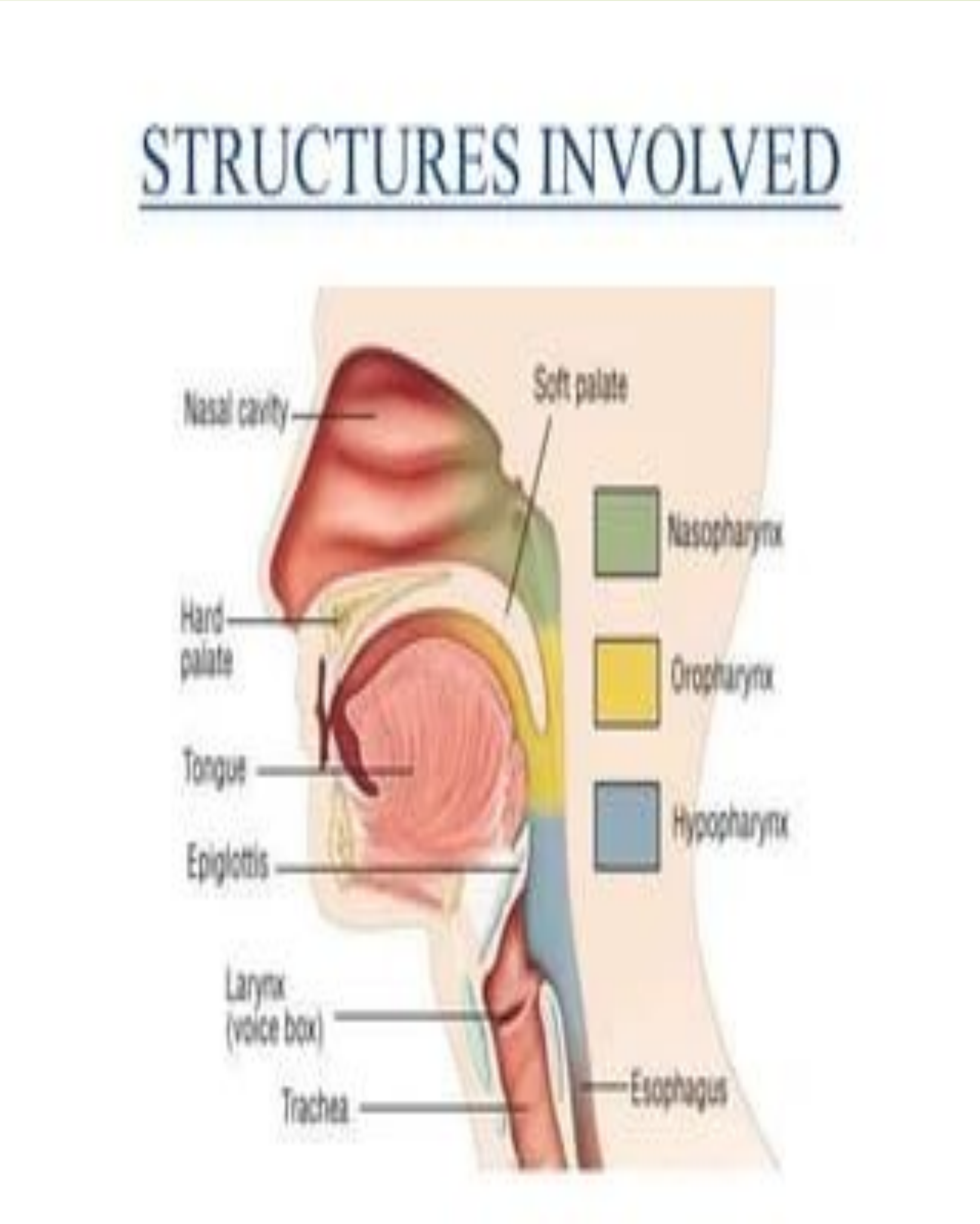
By the end of this lecture students should be able to:

- Define deglutition & identify its phases.
- Describe the three phases of deglutition(oral phase, pharyngeal phase & esophageal phase).
- Differentiate between voluntary & involuntary components of swallowing.
- Identify the sequence of events occurring during oral phase(mastication, bolus formation, propulsion).
- Describe the pharyngeal phase including closure of oropharynx, protection of airway, elevation of larynx, relaxation of UES.
- Explain the esophageal phase including peristalsis.
- Describe the neural control of deglutition & identify the swallowing center & sensory & motor pathway.
- Identify the cranial nerves involved in deglutition & their specific roles.
- Correlate abnormalities with clinical conditions(e.g. dysphagia, achalasia).

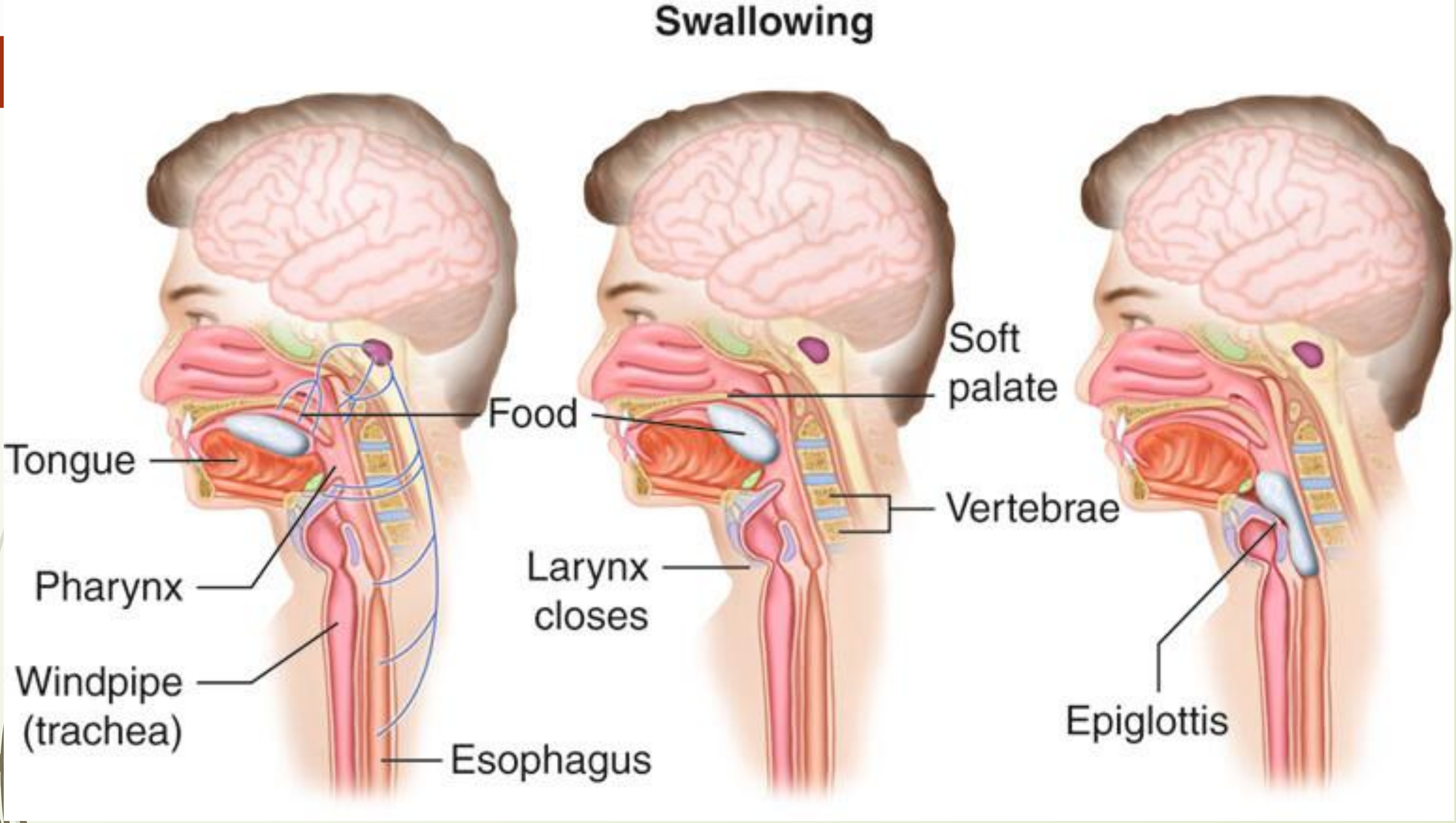
DEGLUTITION

Deglutition is a complex physiological process of swallowing that transports food or liquid from the mouth to the stomach.

- It is a complex neuromuscular process & involves both voluntary & involuntary actions.
- The process of deglutition relies on the highly coordinated action of over 30 muscles & 5 cranial nerves.
- This vital process ensures safely transport of food to the stomach & protection of respiratory airway.
- The whole process of swallowing roughly takes 4 to 8 seconds for solids & only 1 second for liquids.




Swallowing Mechanism
1. Oral Phase
2. Pharyngeal Phase
3. Esophageal Phase

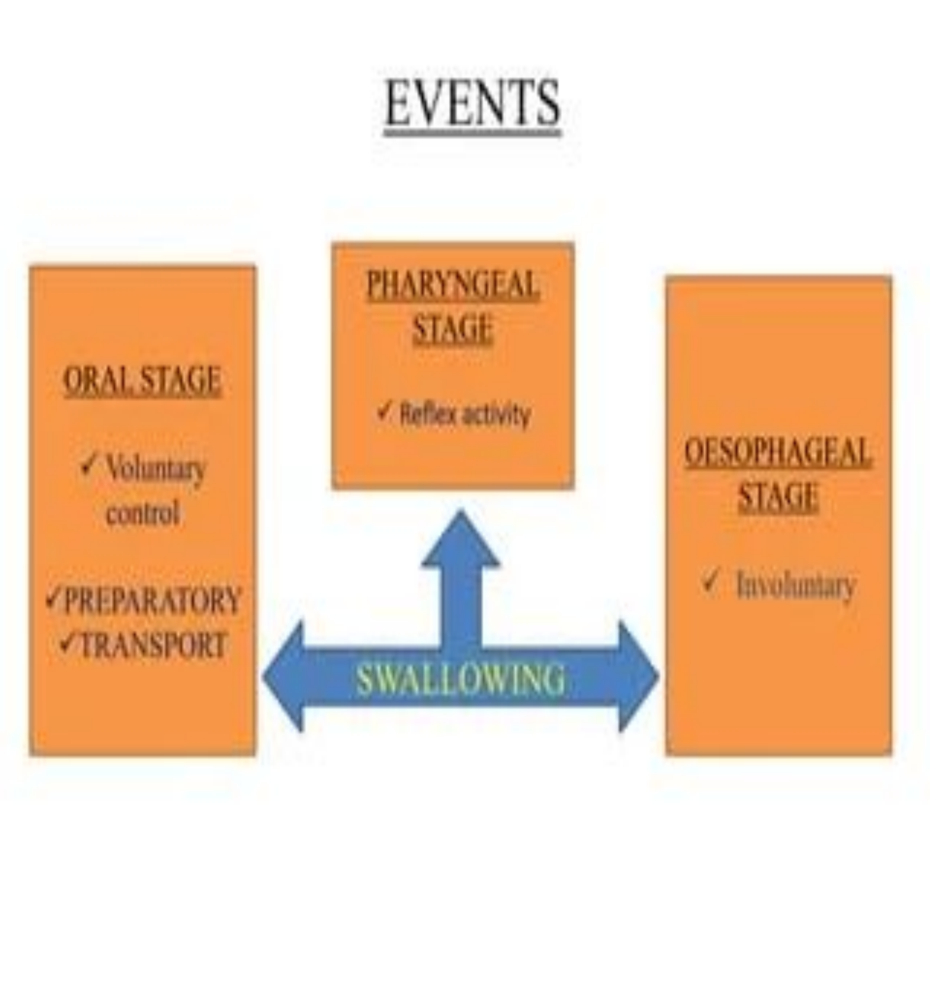


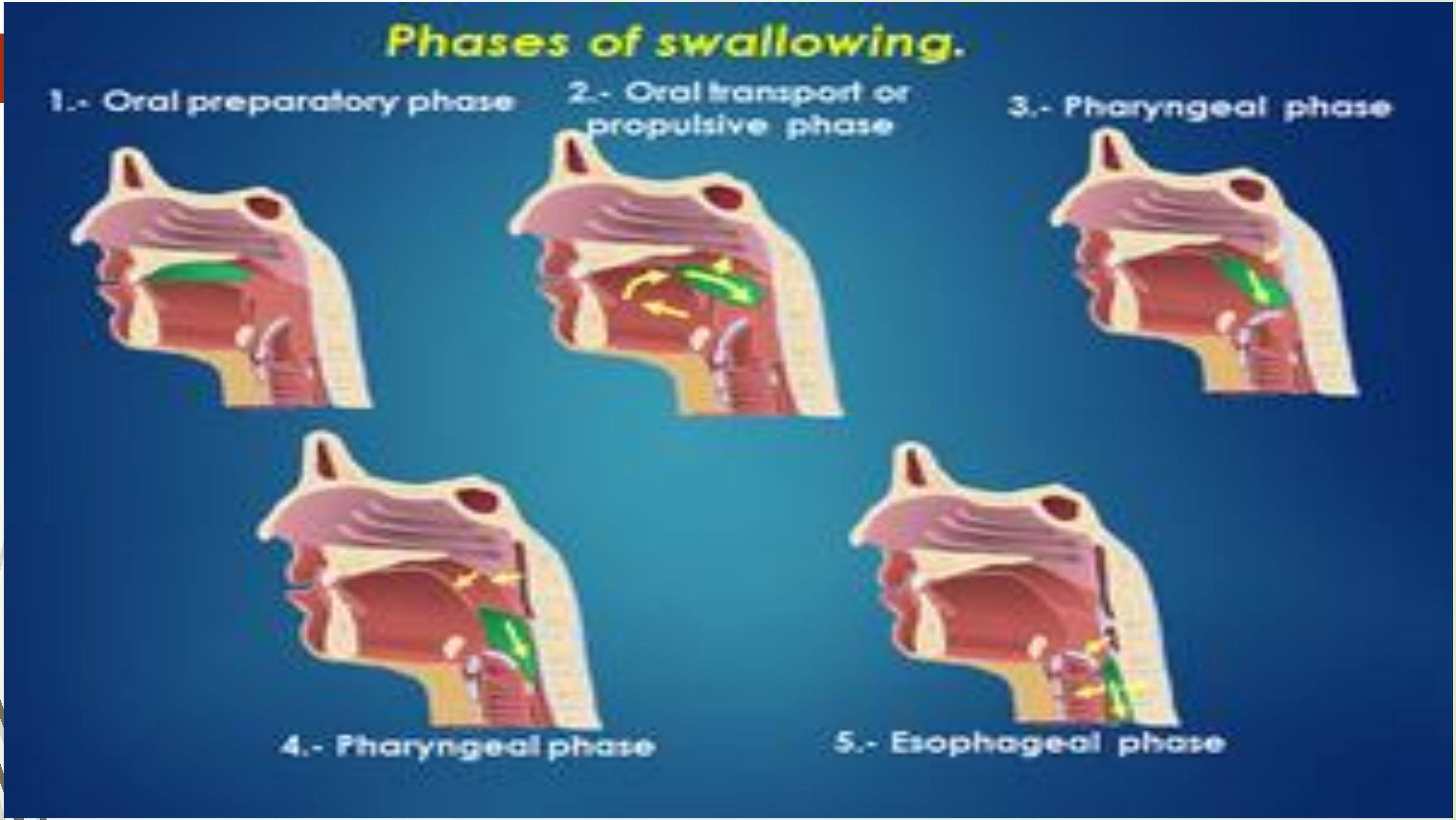
Swallowing
Food
Tongue
Pharynx
Windpipe
(trachea)
Esophagus
Soft
palate
Vertebrae
Larynx
closes
Epiglottis

PHASES OF DEGLUTITION

- Swallowing or deglutition is a fundamental component of digestion.
- It allows food to be safely transferred from the oral cavity to the stomach via esophagus.
- The swallowing mechanism occurs sequentially in three primary phases:

1. *Oral phase(voluntary phase)*
2. *Pharyngeal phase(involuntary phase)*
3. *Esophagal phase(involuntary phase)*




Phases of swallowing.
1.- Oral preparatory phase
2.- Oral transport or propulsive phase
3.- Pharyngeal phase
4.- Pharyngeal phase
5.- Esophageal phase

1. ORAL PHASE

This phase is under conscious control & can be divided into following stages:

Mastication

It involves chewing & mixing of food with saliva to form a cohesive & lubricated ball called ***bolus***.

Positioning

- During this stage, the bolus is retained within the oral cavity by the elevation of tongue against the soft palate.
- Positioning traps the bolus & prevents the premature entry of bolus into the oropharynx while the airway remains open.

CONT..

Propulsion

During this stage, the tongue presses sequentially against the hard palate from front to back & propels the food bolus into the oropharynx.

Trigger

As the bolus hits pressure receptors at the back of mouth, it triggers the automatic swallowing reflex.

- This marks the transition of oral phase to the pharyngeal phase of swallowing.
- The duration of the oral phase varies depending on the food texture & conscious control.

2. PHARYNGEAL PHASE

- This is the second stage of swallowing & is under involuntary control.
- It is a rapid reflex & takes less than a second.
- It begins when the food bolus enters the oropharynx & stimulates pressure receptors in the soft palate & anterior pharynx.
- Stimulation of these pressure receptors activates the swallowing center in the brainstem.
- This triggers a series of coordinated actions to protect the airway & facilitate bolus passage.
- It involves the following stages:

1) Nasal protection
2) Airway protection
3) Bolus passage

Nasal protection

The soft palate & uvula elevates to seal off the nasopharynx & preventing the food from travelling up into the nasal cavity.

Airway protection

- It causes inhibition of respiration i.e. the breathing temporarily stops called ***swallowing apnea***.
- The larynx(voice box) elevates & moves forward causing the epiglottis to fold down over the glottis
- The vocal cords tightly close together to form a barrier against choking(aspiration).

2. Pharyngeal phase of swallowing

Bolus reaches posterior part of mouth & pharynx

stimulates receptors (especially on tonsillar pillars)

Deglutition or swallowing centre in the medulla & lower Pons

Respiratory centre

deglutition apnea

Motor impulses

cranial nerves

Automatic pharyngeal muscle contraction

Bolus passage

- Pharyngeal constrictor muscles squeeze from top to bottom, propelling the bolus down.
- Consequently, the ***upper esophageal sphincter(UES)*** relaxed & opens to receive the food.
- The speed of bolus movement is determined primarily by the viscosity & volume of bolus & only minimally by the role of gravity.

3. ESOPHAGEAL PHASE

- This is the final stage of swallowing & is under involuntary control.
- Once the bolus clears the throat, the larynx drops back down, the epiglottis reopens & normal breathing is resumed.

Regions of esophagus

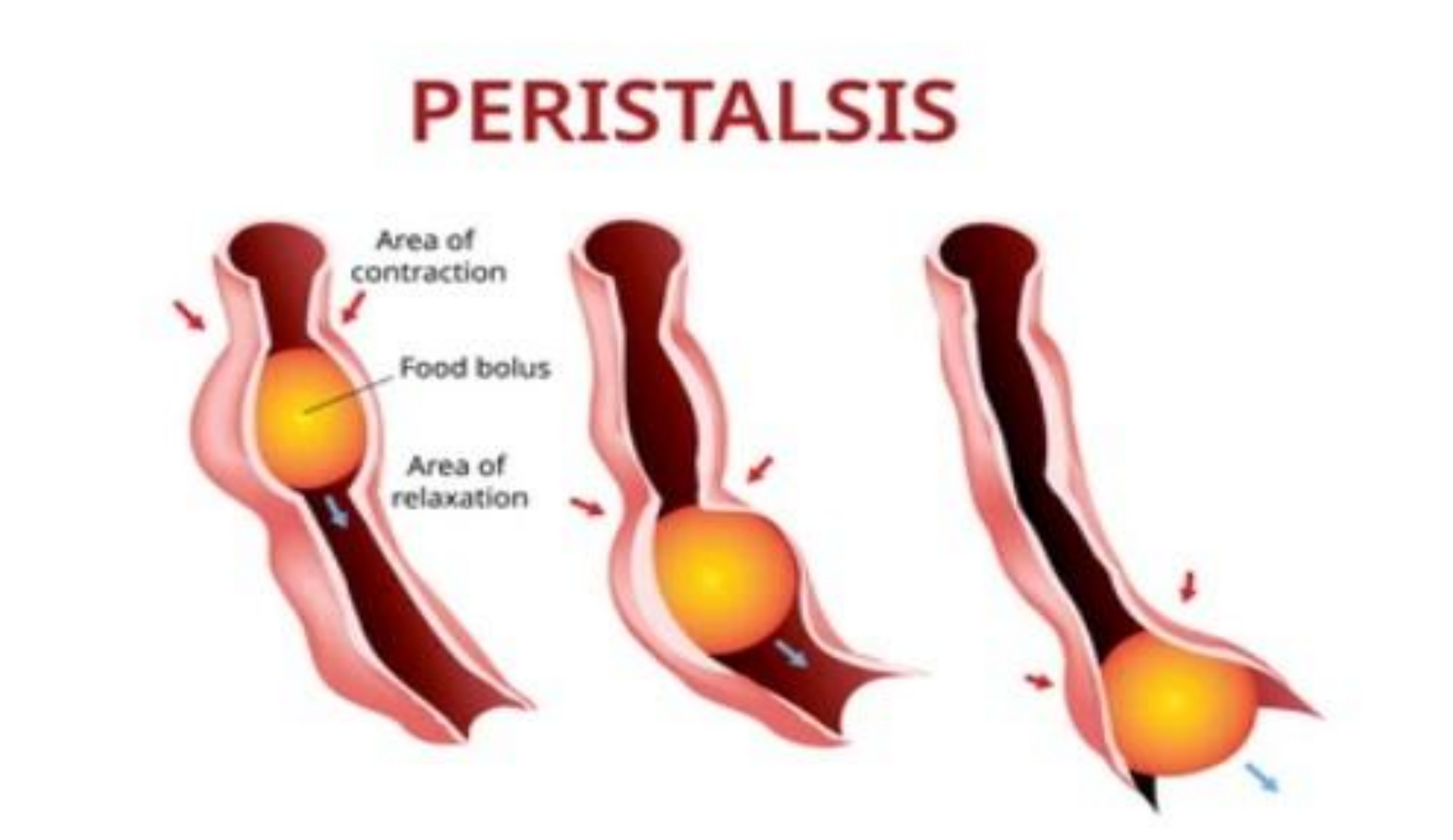
The esophagus is composed of three regions:

- The upper third contains skeletal(voluntary) muscles.
- The lower third contains smooth(involuntary) muscles.
- The middle third contains mixture of both voluntary & involuntary muscles.

Peristalsis

Rhythmic wave like contractions of esophageal smooth & skeletal muscles is termed as peristalsis.

- The bolus is subsequently propelled down the esophagus to towards the stomach via peristalsis.
- Each segment of muscle systemtically relaxes ahead of the bolus to allow passage then contracts behind it to move it onward.


PERISTALSIS
Area of contraction
Food bolus
Area of relaxation

- Peristalsis is primarily controlled by the autonomic nervous system & the bolus transit occurs at approximately 3 to 5 cm/second, taking around 9 seconds to reach the stomach.

Bolus entry into the stomach

- As the bolus reaches the end of line, the lower esophageal sphincter (LES) relaxes to let the food pass into the stomach then clamps shut to prevent acid reflux.

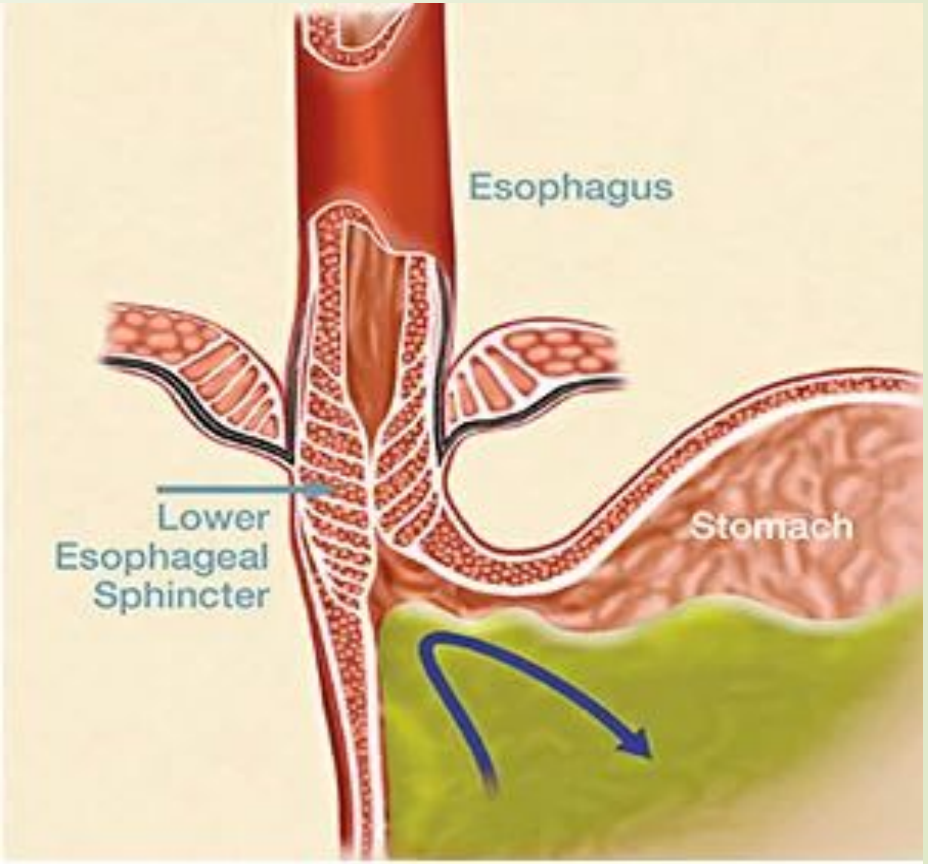


Figure 1: The lower esophageal sphincter (LES) is a muscle located between the stomach and esophagus. Normally, the LES relaxes to allow food and liquid to pass into the stomach and then closes to prevent reflux.

ESOPHAGEAL STAGE

Initiation of peristalsis in esophagus

↓

Bolus is propel down

↓

Bolus enters the lower part of esophagus

↓

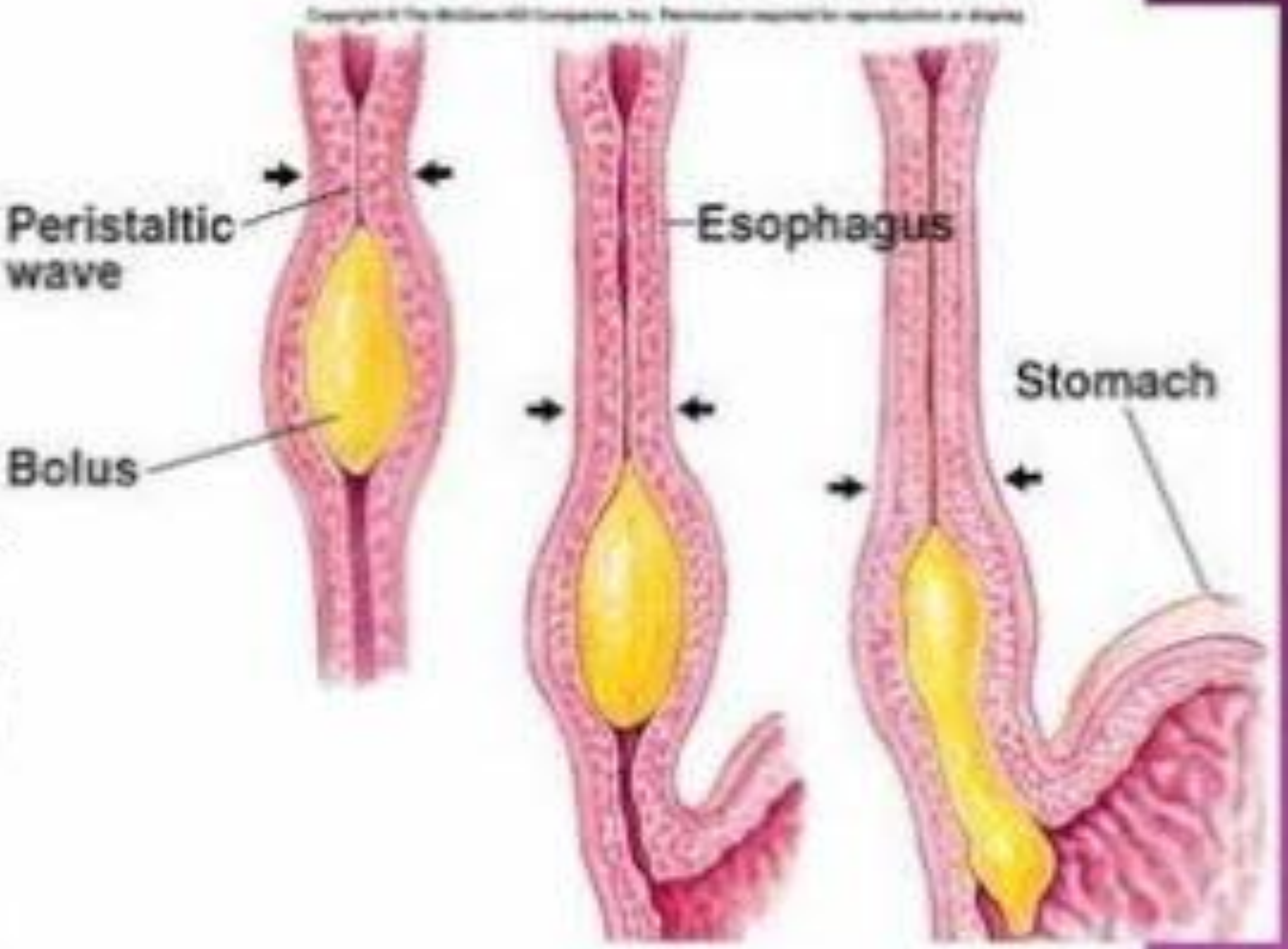
Relaxation of lower esophageal sphincter

↓

Bolus entered to stomach.

ESOPHAGUS

- Peristalsis:
  - Produced by a series of localized reflexes in response to distention of wall by bolus.
- Wave-like muscular contractions:
  - Circular smooth muscle contract behind, relaxes in front of the bolus.
  - Followed by longitudinal contraction (shortening) of smooth muscle.
    - Rate of 2-4 cm/sec.
  - After food passes into stomach, LES constricts.



NEURAL CONTROL OF DEGLUTITION

- The process of deglutition is managed by the ***swallowing center*** in the brainstem(medulla oblongata & lower pons).
- The swallowing center processes sensory signals from the throat & delivers motor commands via 5 cranial nerves:

1. Trigeminal nerve(CN V)

- Trigeminal nerve controls ***muscles of mastication*** to chew the food, the ***mylohyoid muscle***(stabilizes the floor of mouth) & ***tensor veli palatini***(elevates the soft palate).
- It also provides sensory feedback for the anterior two thirds of tongue.

2. Facial nerve(CN VII)

- Facial nerve directs the lips & cheeks(***orbicularis oris***) to keep food inside the oral cavity by managing the ***lip seal*** & prevent drooling.
- It also innervates the submandibular & sublingual salivary glands.
- It provides sensory taste function to the anterior two-third of the tongue.

3. Glossopharyngeal nerve(CN IX)

- Glossopharyngeal nerve provides sensation to the posterior one-third of tongue & ***oropharynx***(important for ***swallowing reflex initiation***).
- It also controls the parotid glands for salivation.

4. Vagus nerve(CN X)

- Vagus nerve manages the motor contraction of ***pharyngeal constrictors***, lifting the soft palate by ***levator veli palatini*** & protecting the airway by closing the glottis.
- It also mediates esophageal motility.

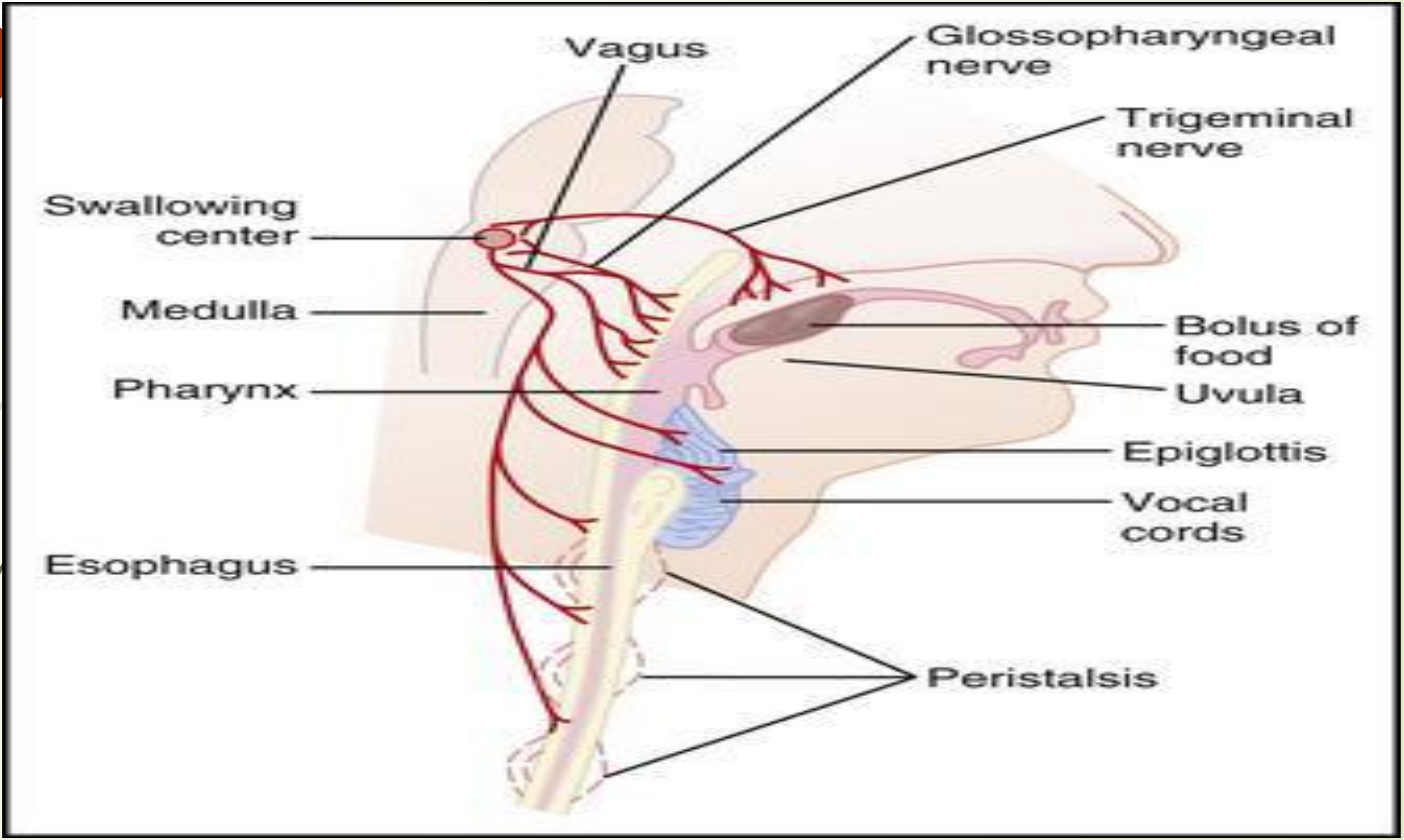
5. Hypoglossal nerve(CN XII)

- Hypoglossal nerve drives all the complex movements of the tongue(intrinsic/extrinsic muscles) necessary for preparing the bolus, manipulating it & propelling the bolus into the pharynx.

Table 1 **Cranial Nerves Involved in Chewing and Swallowing**

Cranial Nerve Number	Name	Function
V	Trigeminal	Facial sensation, biting, and chewing
VII	Facial	Perceiving taste in the front two-thirds of the tongue and production of saliva
IX	Glossopharyngeal	Swallowing and the gag reflex
X	Vagus	Swallowing and the gag reflex
XII	Hypoglossal	Tongue movement

— SOURCE: CICHERO J, MURDOCH BE, EDS. ***DYSPHAGIA: FOUNDATION, THEORY AND PRACTICE***. 1ST ED. WEST SUSSEX, ENGLAND: WILEY; 2006:157-161.


Vagus
Glossopharyngeal nerve
Trigeminal nerve
Swallowing center
Medulla
Pharynx
Esophagus
Bolus of food
Uvula
Epiglottis
Vocal cords
Peristalsis

SWALLOWING CENTER

- The swallowing center is located in the medulla oblongata & lower pons of the brainstem.
- It acts as central coordinator for deglutition, transforming voluntary oral movements into involuntary pharyngeal & esophageal reflexes.
- It processes the sensory input to trigger the swallowing reflex, coordinating over 30 muscles to move food while protecting the airway.

COMPONENTS OF SWALLOWING CENTER

Nucleus tractus solitarius(NTS)

- It is located in the dorsal medulla & acts as the ***receptor area*** which receives sensory input to trigger the swallowing process.

Nucleus ambiguous(NA)

- It is located in the ventral medulla & acts as the ***motor area*** issuing commands to the muscles of the pharynx & esophagus.

FUNCTIONS OF SWALLOWING CENTER

1. ***Triggering the reflex:***

- It receives sensory input via cranial nerves V, IX & X from the posterior mouth & oropharynx regarding the arrival of bolus.

2. Sequential coordination:

- It triggers the involuntary pharyngeal phase, coordinating the rapid contraction of pharyngeal muscles & ensuring the larynx to elevate & close to prevent aspiration.

3. Airway protection:

- The swallowing center inhibits respiratory centers in the brainstem, stopping breathing briefly to prevent food from entering the trachea.

4. Muscles sequencing:

- The swallowing center coordinates the opening of the upper esophageal sphincter to allow the bolus to pass & initiates esophageal peristalsis.

DEGLUTITION REFLEX

- Although the beginning of swallowing is a voluntary act, however later it becomes involuntary & is carried out by a reflex action called ***deglutition reflex***.
- Deglutition reflex occurs in the pharyngeal & esophageal stages of swallowing.

STIMULUS

- As the bolus enters the oropharyngeal region, the receptors present in this region are stimulated.

AFFERENT PATHWAY

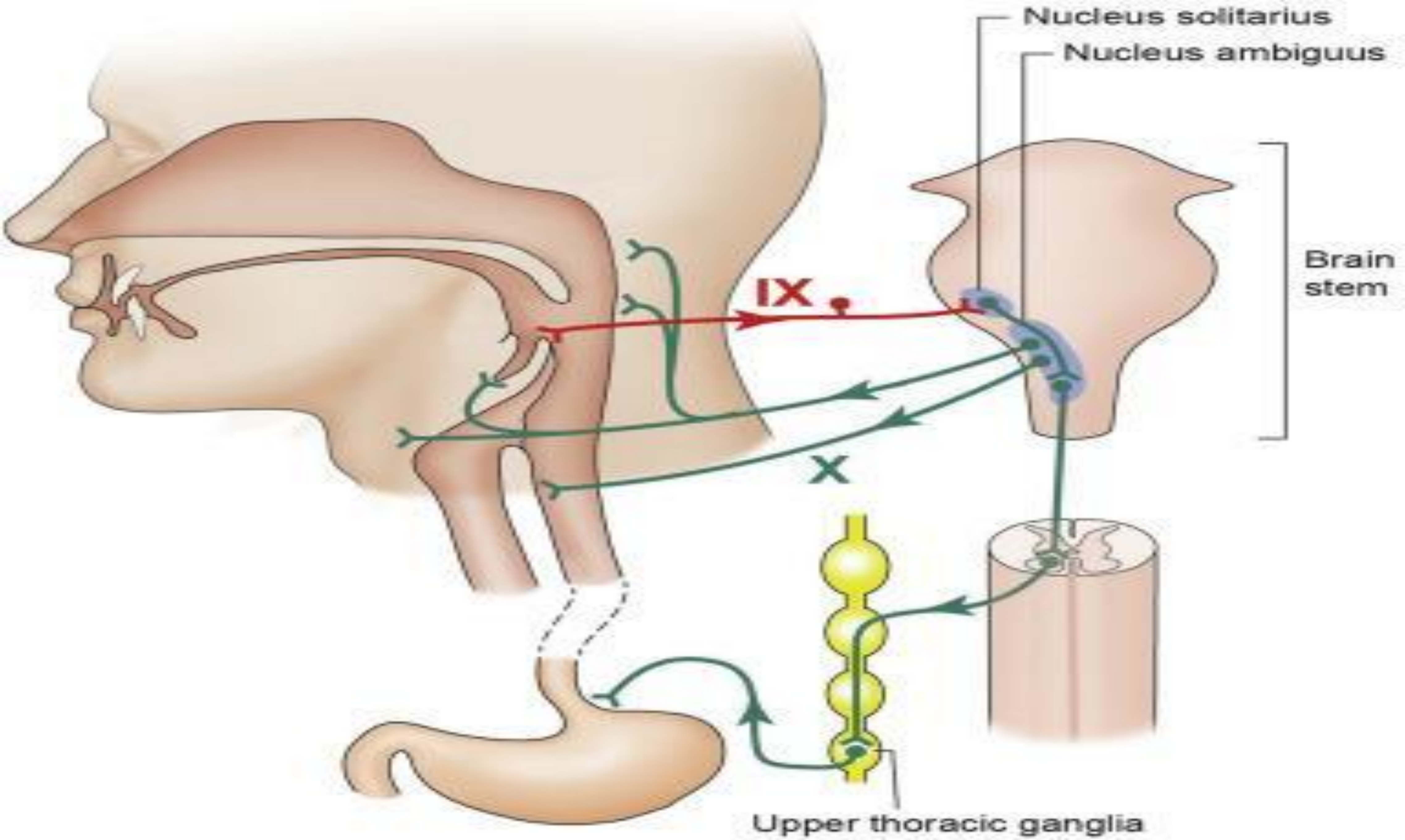
- Afferent impulses from the oropharyngeal receptors pass via ***glossopharyngeal nerve*** fibres to the deglutition center.

EFFERENT PATHWAY

- Impulses from the deglutition center travel through the ***glossopharyngeal & vagus nerves***(parasympathetic motor fibers) & reach the soft palate, pharynx & esophagus.
- The glossopharyngeal nerve is concerned with the pharyngeal phase of swallowing.
- The vagus nerve is concerned with the esophageal phase.

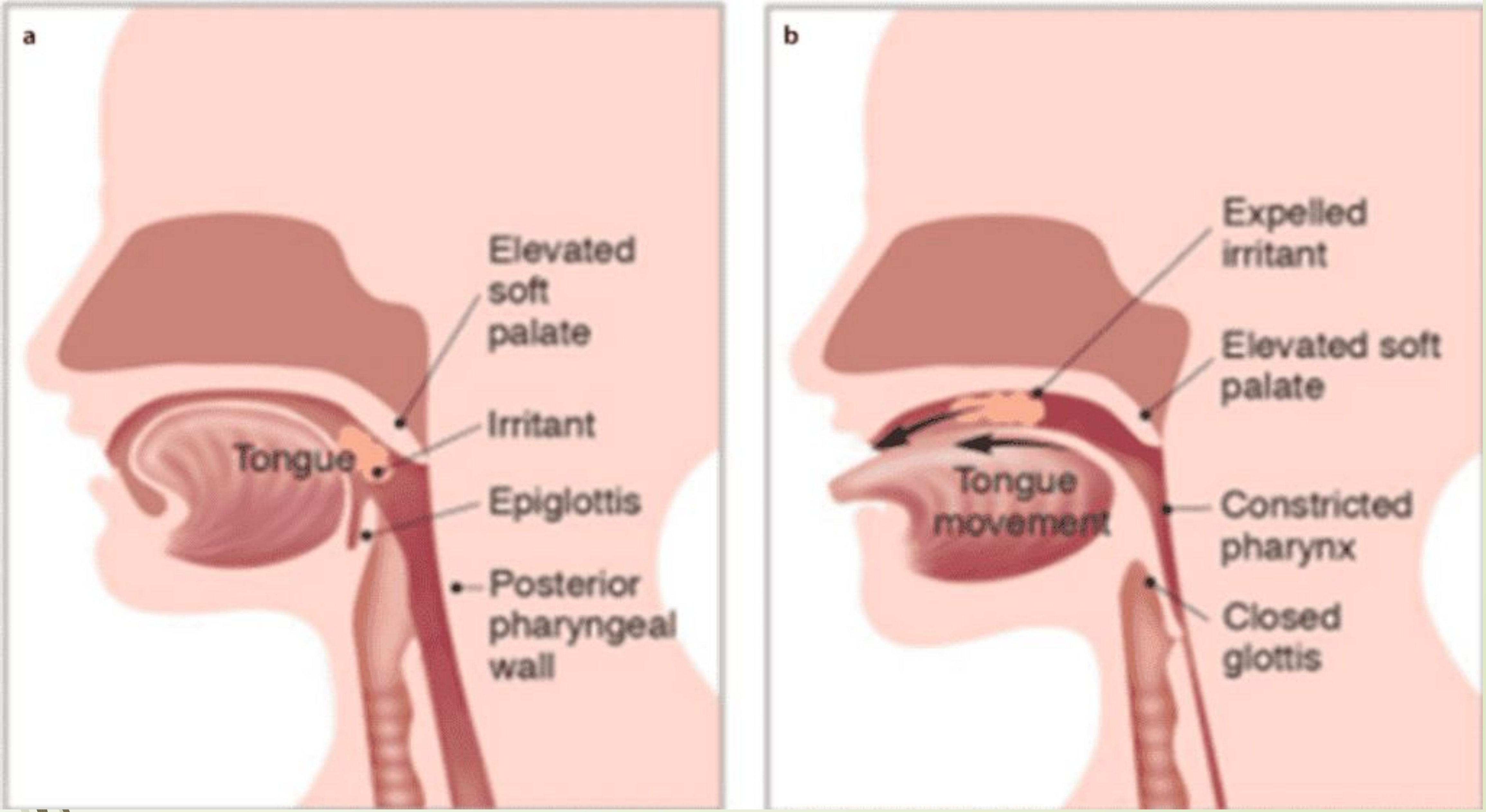
RESPONSE OF DEGLUTITION REFLEX

- The swallowing reflex causes upward movement of soft palate to close the nasopharynx & upward movement of larynx to close the respiratory passage so that the bolus enters the esophagus.
- Now the peristalsis occurs in the esophagus which pushes the bolus into stomach.


Nucleus solitarius
Nucleus ambiguus
Brain stem
IX
X
Upper thoracic ganglia

GAG REFLEX

- Swallowing hard & solid food in babies below 6 months is prevented by gag reflex.
- Gag reflex also known as pharyngeal reflex is a normal protective reflex consisting of strong involuntary effort to vomit(retching).
- This reflex prevents choking by thrusting objects in the throat towards the opening of the mouth.
- It is initiated by touch of an object at the roof of the mouth, back of the tongue, the area around the tonsils & the back of throat.
- Gag reflex is very active in babies below 6 months to prevent the passage of hard & solid food into the throat.
- After 6 months, the gag reflex diminishes enabling the baby to swallow chunky & solid food.
- It occurs by the mass contraction of posterior oral & pharyngeal muscles & elevation of the soft palate.
- Sensory information is carried by the glossopharyngeal nerve & motor information is carried by the vagus nerve.
- Center for gag reflex is in the nuclei of X cranial nerve(vagus nerve).


a
Elevated
soft
palate
Irritant
Tongue
Epiglottis
Posterior
pharyngeal
wall
b
Expelled
irritant
Elevated soft
palate
Tongue
movement
Constricted
pharynx
Closed
glottis

EFFECTS OF CRANIAL NERVE DAMAGE ON DEGLUTITION

DAMAGE TO CN V

- Loss of oral sensation leads to reduced bolus control & awareness of bolus location.
- Reduced mastication skills due to damage to the mandibular branch of CN V.
- Reduced airway protection increases the risk of aspiration due to compromised epiglottic down folding & UES opening.
- Increased risk of nasophsryngeal reflux.

DAMAGE TO CN VII

- Reduced oral containment & drooling due to reduced lip function.
- Reduced taste sensation in anterior two-third of tongue.
- Compromised function of submandibular & sublingual salivary glands resulting in decreased oral saliva production.

DAMAGE TO CN IX

- Reduced sensation on posterior one third of tongue & oropharynx causing altered taste & reduced trigger response.
- Reduced gag reflex.
- Poor posterior oral containment due to reduced posterior oral sensation.

DAMAGE TO CN X

- Reduced laryngeal valving due to damage to innervation of intrinsic laryngeal muscles causing increased risk of aspiration during swallowing.
- Reduced sensory function to the pharynx & larynx results in reduced reflexive cough response which increases the risk of aspiration.
- Impaired upper esophageal sphincter(UES) opening.

DAMAGE TO CN XII

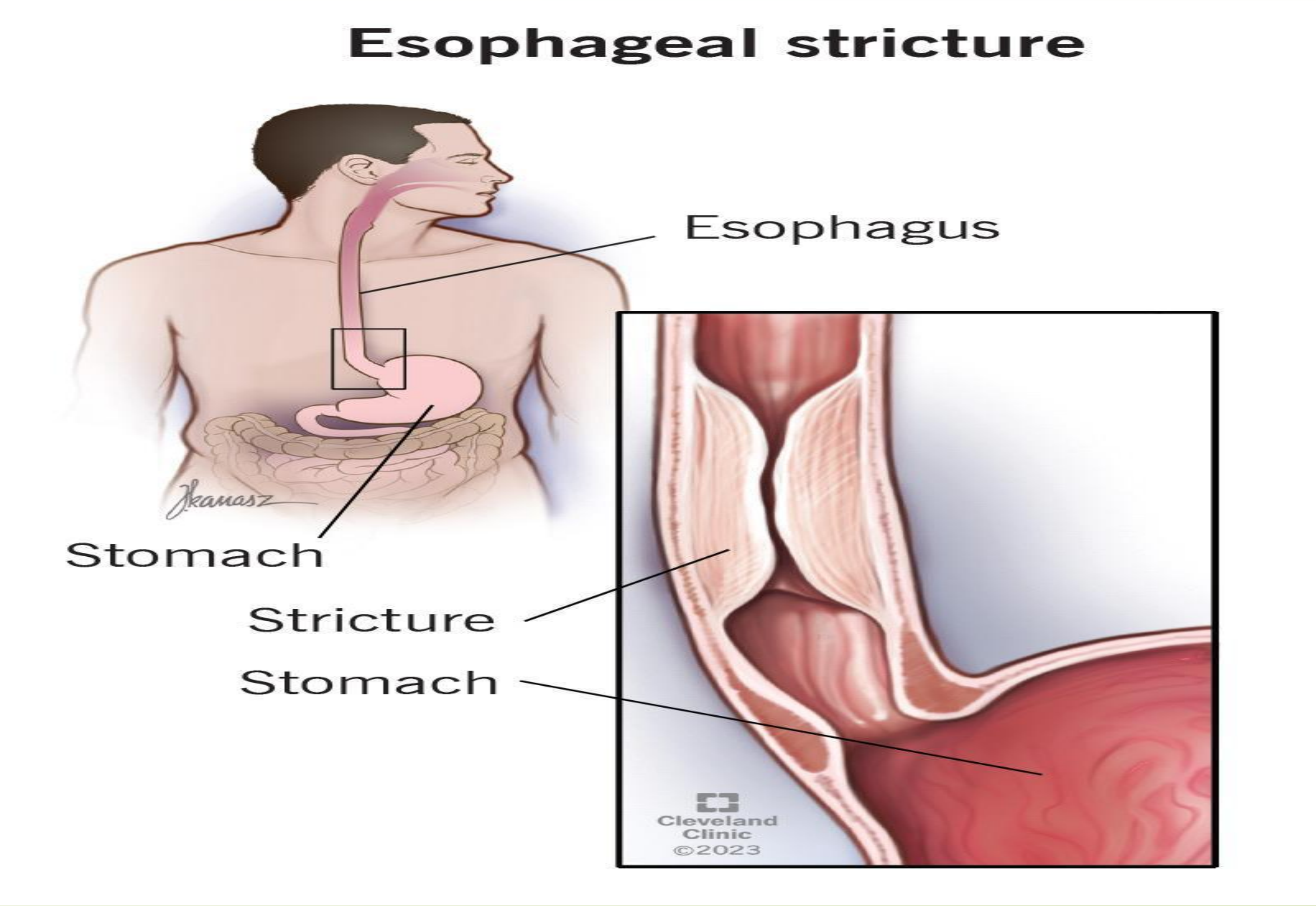
- Poor bolus control in the oral cavity leading to reduced mastication & potential residue in oral spaces after swallowing due to compromised innervation of intrinsic & extrinsic muscles of tongue.

APPLIED PHYSIOLOGY

1. DYSPHAGIA

- Difficulty in swallowing is termed as dysphagia.
- Causes include:

1) Mechanical obstruction of esophagus due to tumor, stricture(narrowing of tubular structure), diverticular hernia etc.
2) Decreased movement of esophagus due to neurological disorders such as ***parkinsonism***.
3) Muscular disorders leading to difficulty in swallowing during oral stage or esophagal stage.


Esophageal stricture
Esophagus
Stomach
Stricture
Stomach
Cleveland Clinic
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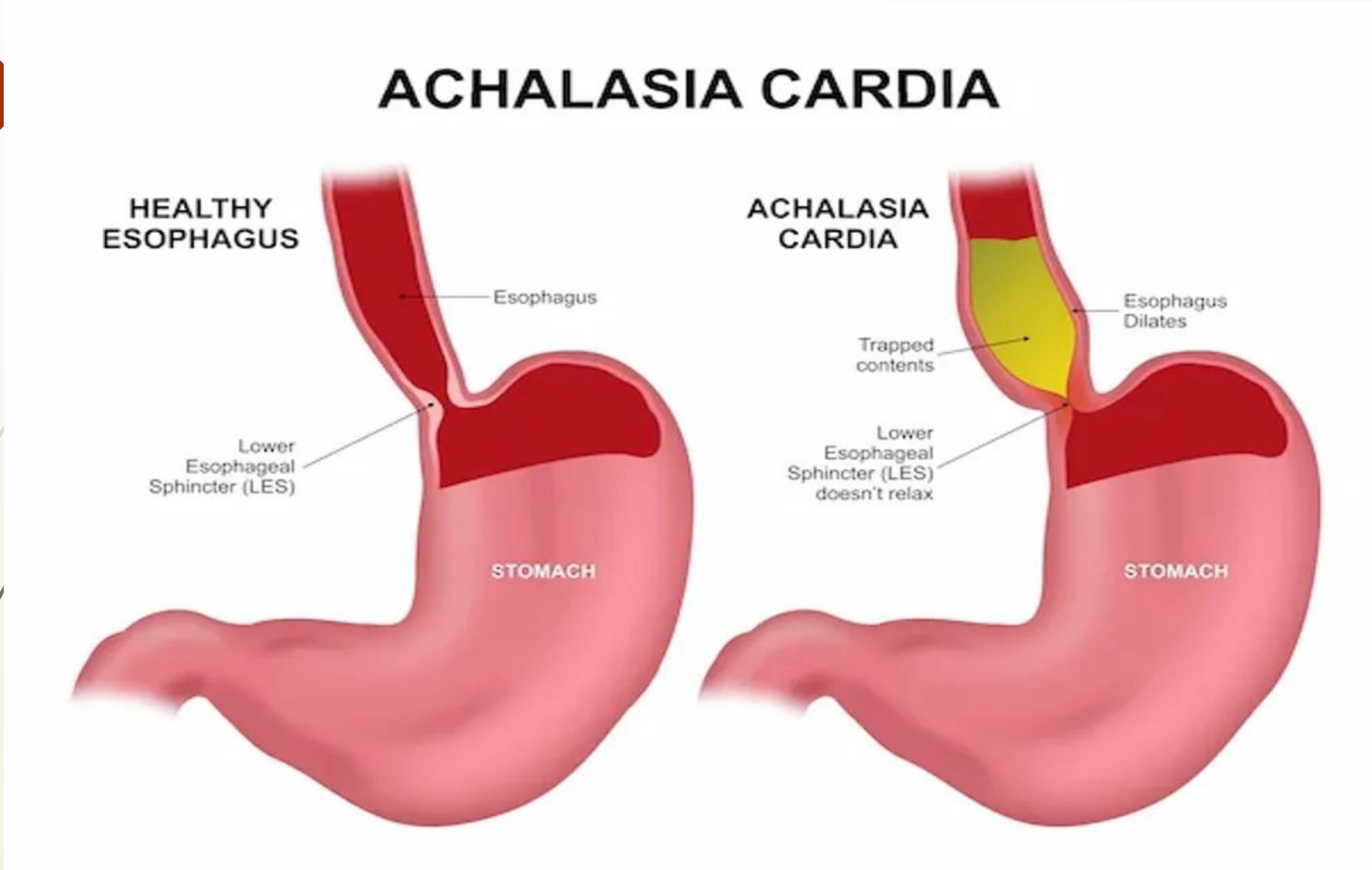
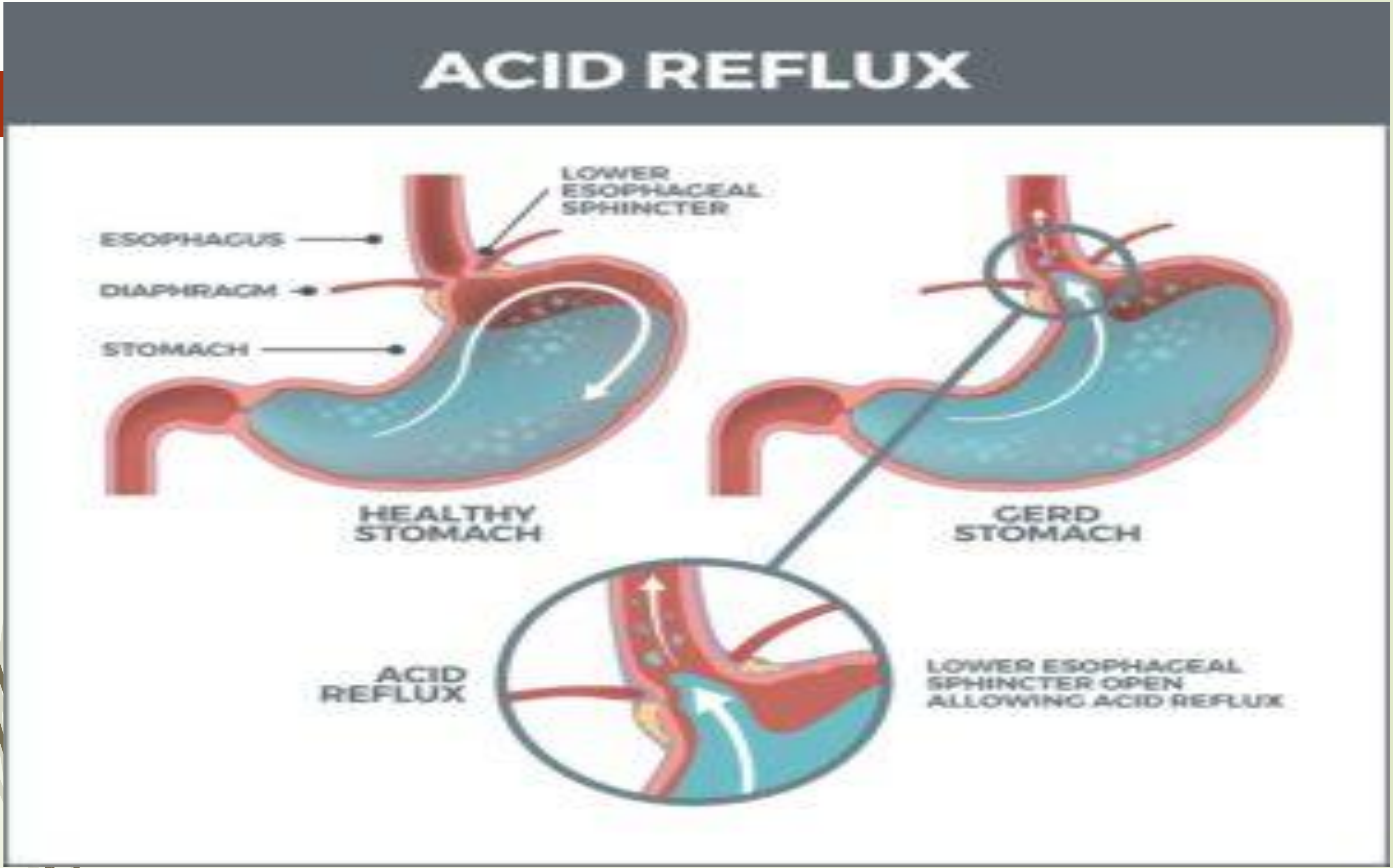
2. ESOPHAGEAL ACHALASIA

- It is a neuromuscular disease characterized by the accumulation of food substances in the esophagus that inhibits normal swallowing.
- It is due to the failure of lower esophageal sphincter(LES) to relax during swallowing.
- The accumulated food substances cause dilation of esophagus.
- Features of esophageal achalasia include dysphagia, chest pain, weight loss & cough.


ACHALASIA CARDIA
HEALTHY ESOPHAGUS
Esophagus
Lower Esophageal Sphincter (LES)
STOMACH
ACHALASIA CARDIA
Esophagus Dilates
Trapped contents
Lower Esophageal Sphincter (LES) doesn't relax
STOMACH

3. GASTROESOPHAGEAL REFLUX DISEASE(GERD)

- It is a condition in which stomach acid & contents flow back into the esophagus.
- It occurs when lower esophageal sphincters(LES) weakens & relaxes improperly allowing stomach contents to escape upward.
- Most widely reported symptoms of GERD include heartburn(painful burning sensation in the chest), regurgitation(feeling of bitter or sour tasting acid coming up into the throat or mouth), dysphagia(difficulty in swallowing) & bad breath.
- Risk factors include obesity & excess abdominal weight, pregnancy, smoking & hiatal hernia(a condition in which part of stomach protrudes through the diaphragm into the chest cavity).


ACID REFLUX
LOWER ESOPHAGEAL SPHINCTER
ESOPHAGUS
DIAPHRAGM
STOMACH
HEALTHY STOMACH
GERD STOMACH
ACID REFLUX
LOWER ESOPHAGEAL SPHINCTER OPEN ALLOWING ACID REFLUX

POST ASSESSMENT

Q1. The swallowing center is located in the:

a) Cerebellum
b) Midbrain
c) Medulla oblongata
d) Hypothalamus

Q2. Which structure closes the laryngeal opening during swallowing:

a) Uvula
b) Vocal cords
c) Tongue
d) epiglottis

Q3. The pharyngeal phase of swallowing is:

a) Completely voluntary
b) Controlled by spinal cord
c) Reflex & involuntary
d) Under hormonal control

Q4. The main defect in achalasia is:

a) Increased gastric acid secretion
b) Weak contraction of pyloric sphincter
c) Excess salivation
d) Failure of lower esophageal sphincter relaxation